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Workflow Automation: Practical Course Curriculum

Cypher Analytica — From repetitive tasks to automated business processes

A practical course that teaches students to connect applications, move and transform data, apply business rules, and build reliable workflows.

Duration: 40 hours of guided training + 20 hours of assignments and project work
Audience: Students, working professionals, business analysts, entrepreneurs, and operations teams
Prerequisites: Basic computer skills and familiarity with spreadsheets. Programming knowledge is optional.

By the end of the course, learners should be able to design, build, test, document, and monitor a complete business workflow.

Module-wise curriculum

Module	Hours	Topics covered	Practical activity
1. Introduction to Workflow Automation	3	What automation means; manual versus automated processes; triggers, actions, conditions, and outputs; scheduled and event-driven workflows; suitable and unsuitable automation tasks	Identify five repetitive tasks and select one suitable for automation
2. Process Mapping and Workflow Design	3	Process discovery; flowcharts; inputs and outputs; dependencies; business rules; exceptions; identifying human approval points	Map a student enquiry-to-enrolment process
3. Getting Started with an Automation Platform	4	Workspace setup; connectors and nodes; credentials; manual, scheduled, and application triggers; field mapping; testing individual steps	Build a form submission → spreadsheet → notification workflow
4. Working with Data	4	Data types; JSON basics; extracting fields; cleaning text; date and time handling; expressions; missing values; filtering and transforming records	Clean enquiry data, standardise phone numbers, and flag incomplete records

Module	Hours	Topics covered	Practical activity
5. Conditions, Branching, and Iteration	4	IF/ELSE logic; multiple routes; filters; loops; batching; merging results; delays; reusable workflow components	Route enquiries by course interest and assign follow-up priority
6. APIs, Webhooks, and Authentication	5	API fundamentals; HTTP methods; endpoints; headers; query parameters; request bodies; response codes; API keys; OAuth concepts; webhook triggers; polling versus webhooks	Receive a webhook and retrieve information from a test API
7. Business Application Integrations	4	Spreadsheets; email; calendars; cloud storage; CRM concepts; database basics; synchronising records across applications	Automate an assignment submission tracker and reminder process
8. AI-Assisted Workflows	4	Adding an LLM step; prompt templates; summarisation; classification; structured output; validating AI responses; human review; workflow automation versus AI agents	Classify student enquiries and draft replies for approval
9. Reliability, Security, and Monitoring	4	Error handling; retries and backoff; timeouts; rate limits; duplicate prevention; safe reruns; logs; alerts; permissions; protecting credentials and personal data	Simulate a failed integration and recover without creating duplicate records
10. Capstone Project and Deployment	5	Requirements; end-to-end construction; test cases; deployment; monitoring; documentation; handover; estimating usage costs and time saved	Demonstrate a complete workflow with exception handling and an operating guide
Total	**40**		

Suggested teaching tools

Use **one primary platform** throughout the course so learners gain depth. Possible options include **n8n, Make, or Microsoft Power Automate**, depending on the institute's teaching environment and application access.

Supporting tools can include:

- **Forms and spreadsheets:** Collect inputs and maintain records.
- **Email and calendar applications:** Send notifications and schedule activities.
- **An API testing tool:** Practise requests and inspect responses.
- **A test database:** Practise reading and updating records.
- **An LLM service:** Add optional classification, summarisation, and drafting steps.

Account requirements, connector access, and usage costs should be checked before finalising the lab setup.

Practical assignments

1. **Enquiry capture:** Save form responses, validate required fields, and send an acknowledgement.
2. **Lead routing:** Assign enquiries based on course, location, or preferred batch.
3. **Assignment reminders:** Identify pending submissions and prepare reminders without sending duplicates.
4. **Report automation:** Aggregate sample data and generate a scheduled summary.
5. **Approval workflow:** Route a request to a reviewer and record the decision.
6. **AI-assisted classification:** Categorise incoming messages and send uncertain cases for human review.

Capstone: Cypher Analytica Enquiry Management

Learners build a workflow that:

1. Receives a new course enquiry.
2. Validates and standardises the submitted information.
3. Checks whether the enquiry already exists.
4. Creates or updates the enquiry record.

5. Routes it to the relevant counsellor.
6. Sends an approved acknowledgement.
7. Schedules a follow-up reminder.
8. Logs successful actions and alerts an administrator to failures.

Deliverables: Process map, working workflow, test results, setup instructions, and a short demonstration. Use synthetic student data during training.

Assessment and Certification

Assessment	Weight
Practical assignments	30%
Troubleshooting exercise	20%
Capstone project	40%
Demonstration and viva	10%